

# TRiP

Transit **R**idership **P**rediction Tool

## User Guide

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## INTRODUCTION AND BACKGROUND

### PURPOSE

For transit agencies to create budgets and plan for the following year, it is important to have an estimate of the future ridership. The Transit Ridership Prediction Tool was created to assist in this process. The tool is designed for small- to medium-sized transit agencies whose service consists of mostly buses.

### ACKNOWLEDGEMENTS

This project was funded through the TRB IDEA program and was led by the University of Tennessee with support from University of California, Davis and the University of Kentucky. It was advised by a panel of practitioner from WeGo, Unitrans, CDM Smith, and Fehr & Peers.

### HOW IT WORKS

The app requires internal data from the transit agency and external data that is programmed into the app. Using this data, a fixed effects regression model is created using a backwards stepwise approach. After inputting the estimated changes to each variable in the model, the app produces a forecast for the next year's ridership.

## DATA

---

### Data Gathered by User

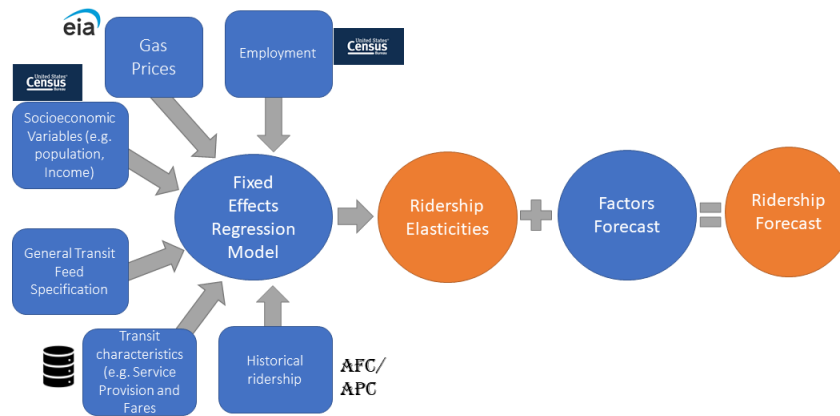
- Total unlinked passenger trips (UPT) per route, per month
- Vehicle revenue miles (VRM) per route on weekdays each month
- General Transit Feed Specification (GTFS) data
- Information on any fare changes in the last 4 years

### Data Provided in the App

- Gas price at the regional level (US Energy Information Administration)
- Total workers 16 years and over (American Community Survey)
- Number of workers working from home (American Community Survey)
- Number of workers commuting by taxicab, motorcycle or other means (American Community Survey)
- Number of workers commuting by car (American Community Survey)
- Number of households with no vehicle available (American Community Survey)
- Number of workers with no vehicle available (American Community Survey)
- Percent of workers who are below 100 percent of the poverty level (American Community Survey)
- Percent of workers who are between 100 to 149 percent of the poverty level (American Community Survey)
- Percent of workers who are female (American Community Survey)
- Percent workers in households in renter-occupied housing units (American Community Survey)
- Total population per census tract (American Community Survey)
- Labor force participation rate per census tract (American Community Survey)
- Unemployment rate per census tract (American Community Survey)

## APP WORKFLOW

The main workflow of the regression model and forecasting process used in the app is shown below.



### Modeling

The model used in the tool is a fixed effect linear regression model with the route being the fixed effect. This helps remove variability associated with context and other unchanging factors related to the route. The basic form is shown below.

Fixed effects model:  $\ln(y_{it}) = \ln(x_{it}) * \beta + \alpha_i EF_i + \rho * Seasonality + \epsilon_{it}$

Where:

$y_{it}$ : ridership for route i during time t (month)

$x_{it}$ : explanatory variables (e.g., service provision) for station i during time t

$EF_i$ : entity fixed effect dummy, equal 1 for route i and 0 otherwise

**Seasonality**: a factor variable to control for the difference in demand by month

$\epsilon_{it}$ : error term

### Forecasting

Once a model is created, the transit agency using the tool must create three scenarios where they give estimates for the percent change of each variable in the final model. Using those estimates, the app creates ridership forecasts for each scenario for the following year.

## APP LAYOUT

The TRiP tool is divided into 8 sections to help users more easily follow the modeling and forecasting process.

1. **Screening:** A few questions to help users determine the usefulness of the app to their agency.
2. **Ridership Data Upload:** Help creating and uploading the proper data for the app to work.
3. **GTFS Upload:** A page to upload GTFS data.
4. **Model Creation:** An interface to create and select a regression model.
5. **Model Review:** A final review of the selected model.

6. **Forecasting Inputs:** A page to forecast ridership based on estimates for three future scenarios.
7. **Visualization:** Selection of plots view forecasted ridership.
8. **Export:** Page to view and download forecast data.

## TAB 1: SCREENING

### OVERVIEW

*This is an initial check to make sure the app is suitable for your transit agency. You will see the first box with some instructions and questions to answer. The second box is empty to start, but some text and a button will appear once you click the “Check Compatibility” button.*

The screenshot displays the 'TRIP App' interface, specifically the '1. Screening' tab. The navigation bar at the top includes: 1. Screening, 2. Ridership Data Upload, 3. GTFS Upload, 4. Model Creation, 5. Model Review, 6. Forecasting Inputs, 7. Visualization, and 8. Export.

The main content area is divided into two panels. The left panel contains introductory text and three screening questions, each with 'Yes' and 'No' radio button options. The right panel shows a green success message and a blue 'Continue' button.

Annotations with red arrows and text identify the following steps:

- Step 1:** Points to the first question: "Has your agency had a system redesign within the last three years?".
- Step 2:** Points to the "Check Compatibility" button at the bottom of the left panel.
- Step 3:** Points to the green message box on the right: "Your responses are compatible with the intended use of this tool."
- Step 4:** Points to the blue "Continue" button on the right panel.

The questions in the left panel are:

- Has your agency had a system redesign within the last three years?  
☐ Yes  
☒ No
- Is a majority of your system ridership from a university or a single employer?  
☐ Yes  
☒ No
- Does your city have light- or heavy-rail transit? If so, have there been any major rail transit investments in the last three years?  
☐ Yes  
☒ No

## STEPS

**STEP 1:** Answer the three questions by selecting the appropriate “Yes” or “No” button.

Has your agency had a system redesign within the last three years?

☐ Yes

☒ No

Is a majority of your system ridership from a university or a single employer?

☐ Yes

☒ No

Does your city have light- or heavy-rail transit? If so, have there been any major rail transit investments in the last three years?

☐ Yes

☒ No

**STEP 2:** Click the blue “Check Compatibility” button. You will see a message appear in the next box.

Check Compatibility

**STEP 3:** Read the message that appears.

**STEP 4:** If you desire to keep using the app, click the blue “Continue” button (it might say “Continue Anyways”). If you don’t think the app will suit your needs, you may close the browser.

Continue

## NOTES

- The app was created to help forecast the next years ridership for small- and medium-sized transit agencies that operate mostly buses.
- While this tab is helpful for first time users, you can skip it by selecting a different tab from the tab selectin pane.

## TAB 2: RIDERSHIP DATA UPLOAD

### OVERVIEW

This is where you will upload your ridership data. It must be an excel file that matches the format specified in the example on this tab. You also have the option to mention any recent fare changes that have occurred.

TRiP App 1. Screening 2. Ridership Data Upload 3. GTFS Upload 4. Model Creation 5. Model Review 6. Forecasting Inputs 7. Visualization 8. Export

**Step 1** Upload a properly formatted excel file with monthly weekday Unlinked Passenger Trips (UPT) and Vehicle Revenue Miles (VRM) for each route. (see example below for data format) Due to unusual ridership patterns during the COVID19 pandemic, it is recommended that your data not go back past the year 2023.

**Step 2** Browse... nashville\_ridership.xlsx Upload complete

**Input Data Example**

| Required |       |      |       |       | Optional |
|----------|-------|------|-------|-------|----------|
| route_id | month | year | upt   | vrn   | is_brt   |
| 14       | 1     | 2025 | 10994 | 18796 | 0        |
| 14       | 2     | 2025 | 10514 | 18822 | 0        |
| 14       | 3     | 2025 | 12670 | 19691 | 0        |
| 14       | 4     | 2025 | 13579 | 18818 | 1        |
| 14       | 5     | 2025 | 13338 | 18796 | 1        |
| 14       | 6     | 2025 | 14453 | 20586 | 1        |
| 14       | 7     | 2025 | 13606 | 17006 | 1        |

**Imported File Preview**

| route_id | month | year | upt      | vrn      |
|----------|-------|------|----------|----------|
| 3        | 1     | 2021 | 14574.41 | 18267.74 |
| 3        | 2     | 2021 | 13770    | 19229.2  |
| 3        | 3     | 2021 | 20699    | 22113.58 |
| 3        | 4     | 2021 | 22566    | 26380.22 |
| 3        | 5     | 2021 | 20694    | 26200.03 |

Showing 1 to 5 of 1,593 Previous 1 2 3 4 5 ... 319 Next

Just one more questions before creating your model for forecasting.

Have you changed your adult base fare in the past 5 years?  
☒ No ☐ Yes

**Step 4**  
**Step 5\***

**Continue**

**Step 6**

\* optional or conditional step

## STEPS

**STEP 1:** Create an excel file with the monthly weekday Unlinked Passenger Trips and Vehicle Revenue Miles for each bus route. As seen in the figure below, the data must contain the following columns: “route\_id”, “month”, “year”, “upt”, “vrn”. If you have routes classified as Bus Rapid Transit, you may optionally add a “brt” column.

| Required |       |      |       |       | Optional |
|----------|-------|------|-------|-------|----------|
| route_id | month | year | upt   | vrn   | is_brt   |
| 14       | 1     | 2025 | 10994 | 18796 | 0        |
| 14       | 2     | 2025 | 10514 | 18822 | 0        |
| 14       | 3     | 2025 | 12670 | 19691 | 0        |
| 14       | 4     | 2025 | 13579 | 18818 | 1        |
| 14       | 5     | 2025 | 13338 | 18796 | 1        |
| 14       | 6     | 2025 | 14453 | 20586 | 1        |
| 14       | 7     | 2025 | 13606 | 17006 | 1        |
| 17       | 1     | 2021 | 3665  | 5408  | 0        |
| 17       | 2     | 2021 | 3378  | 5692  | 0        |
| 17       | 3     | 2021 | 5285  | 6546  | 0        |
| 17       | 4     | 2021 | 5797  | 6932  | 0        |
| 17       | 5     | 2021 | 5207  | 6586  | 0        |
| 17       | 6     | 2021 | 6212  | 7244  | 0        |
| 17       | 7     | 2021 | 6037  | 6915  | 0        |

**STEP 2:** Upload your excel file by clicking the “Browse” button and selecting it from your directory or by dragging and dropping the file into the upload box. The file upload box is found at the top of the first box. If you have not had a fare increase within the last 5 years, you may proceed to Step 7.

Browse...

No file selected

**STEP 3:** A preview of the uploaded file will be found in the second pane (to the right of the screen in most browsers). Review the imported data to make sure it is correct.

| Imported File Preview |       |      |          |          |  |
|-----------------------|-------|------|----------|----------|--|
| route_id              | month | year | upt      | vrn      |  |
| 3                     | 1     | 2021 | 14574.41 | 18267.74 |  |
| 3                     | 2     | 2021 | 13770    | 19229.2  |  |

**STEP 4:** Answer the question regarding fare changes. If you have had a recent change to your adult base fare, click “Yes” and proceed to Step 5. If your adult base fare has not changed in the last 5 years, click “No” and jump to Step 6.

Have you changed your adult base fare in the past 5 years?

☐ No
 ☒ Yes

**STEP 5 (CONDITIONAL):** If you select “Yes”, a few more options will appear in the second pane. Input the date of the fare change, the previous fare (before the fare change), and the new fare (after the change). Once you have filled out each of the three fields, click the “Submit fare increase” button. A table will populate below showing you the data the app is planning on using. You may repeat this if there have been multiple fare changes. Make sure the table accurately reflects the fare change(s). If you notice any mistakes, you can select and delete rows using the “Delete selected fare row” button.

#### Inputting Changes to Adult Base Fare

|                                         |                                   |                                   |
|-----------------------------------------|-----------------------------------|-----------------------------------|
| Date of change                          | Previous fare                     | New fare                          |
| <input type="text" value="2024-06-11"/> | <input type="text" value="2.75"/> | <input type="text" value="3.00"/> |

Submit fare increase

You may select a row from the table below if you want to delete it.

Delete selected row in fare table

| change_date | prev_fare | new_fare |
|-------------|-----------|----------|
| 2023-06-20  | 2.5       | 2.75     |
| 2024-06-11  | 2.75      | 3        |

Continue

**STEP 6:** Once your file is uploaded and any fare changes are properly accounted for, proceed to the next tab by clicking the blue “Continue” button in the second pane or by selecting the appropriate tab in the tab selection pane.

Continue

## NOTES

- The values in the “route\_id” column of your excel file must match the values in the “route\_id” field of the GTFS trips.txt file. (Some agencies use route names that are different from the route IDs listed in the GTFS file)
- If you have your own variables you would also like to put in the model, you may add to your excel file columns that contain positive integers. The model will take the natural logarithm of any added variables, so, if you add any variables, just list actual values and not the logarithm of the values. Also avoid putting 'log' in the column name.
- There will be some warning pop-ups if you have input the wrong type of file.

thly weekday Unlinked

ERROR

You have some unacceptable column names that suggest logarithmic values. The modeling process will take the logarithm of any added variables, so please put original values in any extra columns you add.

Dismiss

thly weekday Unlinked

ERROR

It appears you are missing or have incorrectly spelled some of the required columns (route\_id, month, year, upt, vrm).

Dismiss

thly weekday Unlinked

ERROR

File must be an excel file (.xlsx)

Dismiss

## TAB 3: GTFS UPLOAD

### OVERVIEW

*This is where you will upload your GTFS data. It will show you an interactive map of your uploaded routes, and then it will retrieve census data from the API for the US Census.*

The screenshot shows the '3. GTFS Upload' tab of the TRiP App. The interface includes a progress bar at the top with steps 1 through 8. The main content area is divided into a left sidebar and a right map area. The sidebar contains a 'Route Data Upload' section with a file upload button, a 'GTFS Upload' section with a 'Browse...' button and a file named 'nashville\_gtfs.zip', a text box for instructions, a 'Census API Key' section with a text input and a 'Submit API Key' button, and a 'Get Census Data' button. The map area displays a map of Nashville, Tennessee, with various colored lines representing transit routes. A red box labeled 'Step 1' points to the 'Browse...' button. A red box labeled 'Step 2' points to the text box. A red box labeled 'Step 3' points to the 'Submit API Key' button. A red box labeled 'Step 4' points to the 'Get Census Data' button. A red box labeled 'Step 5' points to a small modal window at the bottom right that says 'Organizing Census Data... Pulling ACS data'.

TRiP App 1. Screening 2. Ridership Data Upload 3. GTFS Upload 4. Model Creation 5. Model Review 6. Forecasting Inputs 7. Visualization 8. Export

Route Data Upload

Upload your agency's GTFS zip file

GTFS Upload

Browse... nashville\_gtfs.zip

Your transit routes and the counties they cross are displayed to the right. If these are the routes and counties you expected, you are ready for the next step. Input your Census API key, and click the button to pull census data.

Census API Key

40-character key

Submit API Key

Get Census Data

Organizing Census Data... Pulling ACS data

Step 1

Step 2

Step 3

Step 4

Step 5

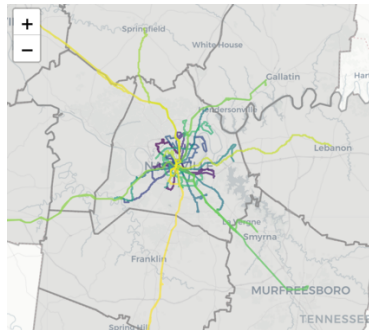
## STEPS

**STEP 1:** Upload your agency's GTFS zip file by clicking the "Browse" button and selecting it from your directory or by dragging and dropping the file into the upload box. The file upload box is found at the top of the first pane.

GTFS Upload

|           |                  |
|-----------|------------------|
| Browse... | No file selected |
|-----------|------------------|

**STEP 2:** An interactive map will be created with the routes from the GTFS file (it may take a few seconds to load the map). Review this to make sure it has the right city and counties you were expecting.



**STEP 3:** Once you know it is the correct data, input your Census API key in the text box and click the "Submit API Key" button.

Census API Key

**STEP 4:** Click the "Get Census Data" button. The app will use your API to gather demographic data around each of your routes using American Community Survey (ACS) data. This data will be used to create additional variables in the regression model. (See next section for details on which variables are used in the model.)

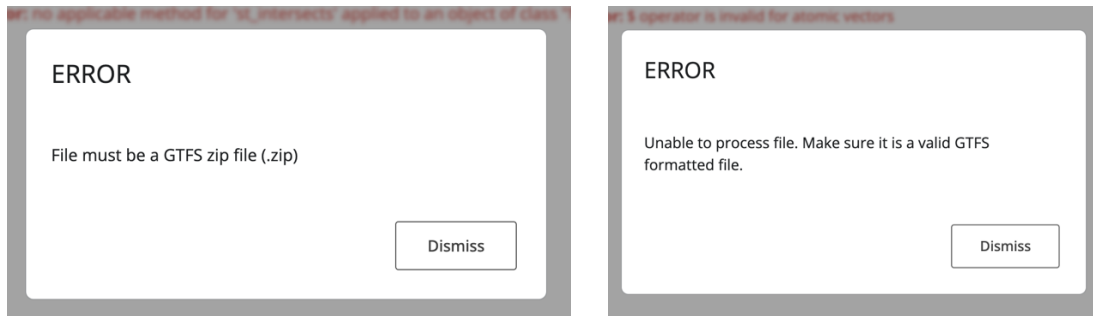
**STEP 5:** Wait for the app to retrieve census data that will be used in your model. Once finished, the app will automatically switch to the next tab.

## NOTES

It is a good idea to use the interactive map to make sure the route IDs in the GTFS match the route IDs in your ridership data.

The app is unable to pull the Census data if the proper ridership data is not uploaded.

There will be some warning pop-ups if you have input the wrong type of file.



## TAB 4: MODEL CREATION

### OVERVIEW

*This is where you will view the regression model that was created and compare it to other models you can optionally create. Then you will choose which model to use for your ridership forecast.*

TRIP App 1. Screening 2. Ridership Data Upload 3. GTFS Upload 4. Model Creation 5. Model Review 6. Forecasting Inputs 7. Visualization 8. Export

Once the data is uploaded, follow these steps to choose a linear regression model to forecast your ridership.

**View Default Model:** The default model will be displayed in the middle of the page. It was generated by the app using a backwards stepwise method (see note below).

**Create Your Own Model:** The selection pane below contains every variable that is available to put in the model (see below for variable descriptions). The pre-selected variables are those belonging to the default model. You may click the selection pane to update which variables you want to test in the model. Note that VRM, Month, and Year variables are required to be selected. Once you have selected variables you want to use in the model, click the **Create New Model** button.

**Variable Selection**

[VRM] [Year] [Year Squared] [Gas Price]  
[% Workers Between 100-150% of Federal Poverty Level]  
[% Commuting by Car] [% No Vehicle Households]  
[% Workers Below Federal Poverty Line] [% Commuting by Taxi]

**Create New Model**

**Default Model Results**

| Variable  | Coeff | Pvalue   |
|-----------|-------|----------|
| VRM       | 0.501 | 0***     |
| February  | 0.109 | 0***     |
| March     | 0.095 | 0***     |
| April     | 0.115 | 0***     |
| May       | 0.077 | 0.004*** |
| June      | 0.070 | 0.038**  |
| July      | 0.102 | 0.003*** |
| August    | 0.281 | 0***     |
| September | 0.335 | 0***     |
| October   | 0.328 | 0***     |
| November  | 0.309 | 0***     |
| December  | 0.220 | 0***     |
| Year      | 0.207 | 0***     |

**Alternative Model Results**

| Variable                                            | Coeff  | Pvalue  |
|-----------------------------------------------------|--------|---------|
| VRM                                                 | 0.542  | 0***    |
| Year                                                | 0.132  | 0***    |
| Year Squared                                        | -0.012 | 0.059*  |
| Gas Price                                           | 0.416  | 0***    |
| % Workers Between 100-150% of Federal Poverty Level | 0.301  | 0.122   |
| % Commuting by Car                                  | -1.493 | 0.041** |
| % No Vehicle Households                             | -0.043 | 0.78    |
| % Workers Below Federal Poverty Line                | -0.223 | 0.112   |
| % Commuting by Taxi                                 | 0.037  | 0.679   |

\* optional or conditional step

Step 1

Step 2\*

Step 3\*

Step 4

## STEPS

**STEP 1:** Review the default model, which is created by the app as soon the ACS data is retrieved. The default model results are displayed in a table in the middle of the screen.

| Variable | Coeff | P.value  |
|----------|-------|----------|
| VRM      | 0.501 | 0***     |
| February | 0.109 | 0***     |
| March    | 0.095 | 0***     |
| April    | 0.115 | 0***     |
| May      | 0.077 | 0.004*** |
| June     | 0.070 | 0.038**  |
| July     | 0.102 | 0.003*** |
| August   | 0.281 | 0***     |

The table reports the coefficient and p-value of each variable, and the rows are highlighted based on the magnitude of the p-value. P-values below 0.1 but above 0.05 are white, p-values below 0.05 but above 0.01 are light green, and p-values below 0.01 are a slightly darker green. Any row with a p-value above 0.1 is highlighted yellow.

For more details on how to interpret the model, see the notes below.

If you are happy with this model and don't want to test any alternative models, you can skip to Step 4.

**STEP 2 (OPTIONAL):** Create a new model by adding or removing variables in the “Variable Selection” input field. To remove variables (e.g. if you want to have a lower p-value threshold), click any variable in the input field and delete it using your computer's delete/backspace key. To add variables, click inside the input field and select from the drop-down any variables you want to add to the model. When you have selected the variables for the new model, click the “Create New Model” button.

Variable Selection

[VRM] [Month] [Year] [Year Squared] [Gas Price]

[% Workers Between 100-150% of Federal Poverty Level] |

[% No Vehicle Households]

[% Workers Below Federal Poverty Line]

[% Commuting by Car]

[% Commuting by Taxi]

[% Work From Home]

Create New Model

**STEP 3 (CONDITIONAL):** Compare the alternative model you created with the default model. On most computer screens, they will be side by side.

| Standard Model Results   |       |          | Alternative Model Results |       |          |
|--------------------------|-------|----------|---------------------------|-------|----------|
| Continue With This Model |       |          | Continue With This Model  |       |          |
| Variable                 | Coeff | P.value  | Variable                  | Coeff | P.value  |
| VRM                      | 0.501 | 0***     | VRM                       | 0.495 | 0***     |
| February                 | 0.109 | 0***     | February                  | 0.109 | 0***     |
| March                    | 0.095 | 0***     | March                     | 0.097 | 0***     |
| April                    | 0.115 | 0***     | April                     | 0.119 | 0***     |
| May                      | 0.077 | 0.004*** | May                       | 0.081 | 0.003*** |
| June                     | 0.070 | 0.038**  | June                      | 0.074 | 0.037**  |
| July                     | 0.102 | 0.003*** | July                      | 0.106 | 0.008*** |
| August                   | 0.281 | 0***     | August                    | 0.287 | 0***     |

**STEP 4:** Once you decide which model you will use to forecast ridership, click on the blue “Continue With This Model” button to proceed to the next step.

Continue With This Model

## NOTES

This “Model Creation” tab will be blank and of no use until you have uploaded the proper files and retrieved the census data.

**Details of the model:** The model used in the tool is a fixed effect linear regression model with the route being the fixed effect. This helps remove variability associated with context and other unchanging factors related to the route. The basic form is shown below.

Fixed effects model:  $\ln(y_{it}) = \ln(x_{it}) * \beta + \alpha_i EF_i + \rho * Seasonality + \epsilon_{it}$

Where:

$y_{it}$ : ridership for route i during time t (month)

$x_{it}$ : explanatory variables (e.g., service provision) for station i during time t

$EF_i$ : entity fixed effect dummy, equal 1 for route i and 0 otherwise

**Seasonality**: a factor variable to control for the difference in demand by month

$\epsilon_{it}$ : error term

**How to interpret the model:** Like any simple linear regression model, a positive coefficient means that the variable is associated with an increase in the dependent variable, and a negative coefficient is associated with a decrease in the dependent variable. The magnitude of the coefficient is associated with the rate that the dependent variable changes. Some examples are shown below.

- -0.3 (fare coefficient): As the fare increase, the predicted passenger trips will decrease.

- 0.5 (VRM coefficient): As the VRM increase, the predicted passenger trips will increase.
- 1.5 (Percent of Low income households): As the Percent of Low income households increase, the predicted passenger trips will increase at a higher rate than with VRM increases.

In this model, the logarithm is taken of the dependent variable and most of the independent variables (with the exception of seasonality variables and the potential brt variable). Taking the logarithm on each side of the equation creates a log-log model which is commonly interpreted as a 1% change in x being associated with a 1% change in y. So, more specifically, the model could be interpreted in the following way.

- -0.3 (fare coefficient): For every 1% increase in fare, the predicted passenger trips will decrease by 0.3%.
- 0.5 (VRM coefficient): For every 1% increase in VRM, the predicted passenger trips will increase by 0.5%.
- 1.5 (Percent of Low income households): For every 1% increase in the Percent of Low income households, the predicted passenger trips will increase by 1.5%.

## TAB 5: MODEL REVIEW

### OVERVIEW

This is where you will review the regression model you have chosen. You have the option to assign different elasticity values for a few of the variables if you determine that would be best.

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**Selected model**

| Variable     | Coeff  |
|--------------|--------|
| VRM          | 0.501  |
| February     | 0.109  |
| March        | 0.095  |
| April        | 0.115  |
| May          | 0.077  |
| June         | 0.070  |
| July         | 0.102  |
| August       | 0.281  |
| September    | 0.335  |
| October      | 0.328  |
| November     | 0.309  |
| December     | 0.220  |
| Year         | 0.207  |
| Year Squared | -0.015 |
| Gas Price    | 0.384  |

W: Median Between 10% SEPM of England

**Pre-forecasting questions**

In the next year, do you plan to convert any of your routes to BRT?  
☒ No ☐ Yes

In the next year, do you plan to increase the adult base fare?  
☒ No ☐ Yes

Research has cited the following elasticities for the coefficients below. If you would prefer to replace or add these coefficients to your model, you can check the corresponding boxes. **Only replace a coefficient if you have good reason to believe the coefficient listed below is a better representation of your local context than the calculated coefficient.**

**Common Coefficient Values**

☐ VRM (0.45)  
☐ Gas Price (0.15)

If you have reviewed the coefficients, added any forced coefficients you wanted, and decided this is the model you want to use, click the button below.

**Proceed to Forecasting**

Step 1

Step 2  
Step 3\*

Step 4

Step 5

\* optional or conditional step

## STEPS

**STEP 1:** Review the model displayed in the first box to make sure it is the one you selected. If there are any coefficient values that seem out of the ordinary, there may be a warning message in the table.

### Selected model

| Variable | Coeff |
|----------|-------|
| VRM      | 0.501 |
| February | 0.109 |
| March    | 0.095 |
| April    | 0.115 |
| May      | 0.077 |

**STEP 2:** Answer the pre-forecasting questions concerning BRT conversion and fare increases.

### Pre-forecasting questions

In the next year, do you plan to convert any of your routes to BRT?

☒ No ☐ Yes

In the next year, do you plan to increase the adult base fare?

☒ No ☐ Yes

**STEP 3 (CONDITIONAL):** If you are planning to convert a route to BRT, then an “Input BRT Changes” section will appear. Using the two input fields. Select the date you will convert a route to BRT and which route(s) will be converted. Make sure the table accurately reflects the BRT change(s). If you notice any mistakes, you can select and delete rows using the “Delete selected row in BRT table” button.

### Input BRT Changes

Please select the route and the approximate date that it will be converted to BRT

Date of change

2027-02-24

Routes converted to BRT

41 28

Submit BRT change

You may select a row from the table below if you want to delete it.

Delete selected row in BRT table

| change_date_brt | routes_brt |
|-----------------|------------|
| 2026-10-11      | 14         |
| 2027-02-24      | 41,28      |

**STEP 4:** Review the common coefficient values in the last pane. You may replace the coefficients of your selected model with any of the options in this pane, but it is also common to leave each of the options unchecked to keep your selected model as-is. Remember to be intentional if you are going to replace or add any coefficients. Only replace a coefficient if you have good reason to believe the coefficient listed is a better representation of your local context than the calculated coefficient in your selected model.

If you check a coefficient that is not already in your model, it will be added to the model without affecting the values of the other coefficients in the model.

#### Common Coefficient Values

- ☐ VRM (0.45)
- ☐ Gas Price (0.15)
- ☐ Fares (-0.3)
- ☐ BRT (0.3)

**STEP 5:** After reviewing the model and making any last tweaks to the coefficients, click the blue “Proceed to Forecasting” button to be directed to the next section.

Proceed to Forecasting

## NOTES

The model shown in this section, along with any changes you make here, is the model that will be used in the forecast.

In the “Input BRT Change” section, you must submit a change for each date you plan to convert a route to BRT, but if multiple routes will be converted to BRT on the same day, you can select multiple routes for that date.

## TAB 6: FORECASTING INPUTS

### OVERVIEW

This is where you will create low, mid, and high scenarios for each route based on how you think the variables might change in the next year. A table will appear previewing all the scenarios before running the ridership forecasts, and you will be updating and saving this table for different routes.

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**Scenario Estimation:** Estimate three scenarios based on realistic assumptions of the direction of each variable (growth/decline) and the magnitude of the expected annual percent change in each variable. For example...

- *Low Estimate* might represent a scenario with small decreases in VRM and other variables.
- *Mid Estimate* might represent a status quo scenario with no changes in VRM and minimal changes in other variables.

Apply scenario to:  
☒ New routes (not yet saved) ☐ Existing routes (overwrite saved routes)

Route  
 All remaining routes

Save

|                                                     | Low Estimate | Mid Estimate | High Estimate |
|-----------------------------------------------------|--------------|--------------|---------------|
| VRM                                                 | -1%          | 2%           | 5%            |
| Gas Price                                           | -1%          | 2%           | 5%            |
| % Workers Between 100-150% of Federal Poverty Level | -1%          | 2%           | 5%            |

Run Forecasts

The assumed forecasts for each route will be displayed below. Check your estimate values once more before proceeding

Show 25 entries Search:

| Route | Variable                                            | Low Estimate | Mid Estimate | High Estimate |
|-------|-----------------------------------------------------|--------------|--------------|---------------|
| 3     | VRM                                                 | -1%          | 2%           | 5%            |
| 3     | Gas Price                                           | -1%          | 2%           | 5%            |
| 3     | % Workers Between 100-150% of Federal Poverty Level | -1%          | 2%           | 5%            |
| 4     | VRM                                                 | -1%          | 2%           | 5%            |
| 4     | Gas Price                                           | -1%          | 2%           | 5%            |
| 4     | % Workers Between 100-150% of Federal Poverty Level | -1%          | 2%           | 5%            |

Step 1: Apply scenario to: ☒ New routes (not yet saved) ☐ Existing routes (overwrite saved routes)

Step 2: Route: All remaining routes

Step 3: Save

Step 4: Table with 4 columns: , Low Estimate, Mid Estimate, High Estimate. Rows: VRM, Gas Price, % Workers Between 100-150% of Federal Poverty Level.

Step 7: Run Forecasts

## STEPS

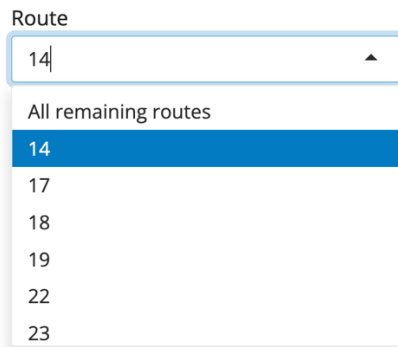
Give more background on how the forecasting works

**STEP 1:** Make sure the “New routes (not yet saved)” option is selected to indicate you are creating the scenario for the first time. Note that the default is to have this option selected.

Apply scenario to:

☒ New routes (not yet saved) ☐ Existing routes (overwrite saved routes)

**STEP 2:** Using the drop-down “Route” selector, choose which route you are creating scenarios for. In the likely case that each route has the same scenario, you will select “All remaining routes.”



**STEP 3:** Double-click in the cells of the table to delete the previewed estimates and type in your agency’s estimated changes (the percent represents how much you expect the value to change by the end of next year).

|                                                     | Low Estimate         | Mid Estimate | High Estimate |
|-----------------------------------------------------|----------------------|--------------|---------------|
| VRM                                                 | <input type="text"/> | 2%           | 5%            |
| Gas Price                                           | -1%                  | 2%           | 5%            |
| % Workers Between 100-150% of Federal Poverty Level | -1%                  | 2%           | 5%            |

**STEP 4:** Once you have filled out the table with estimates for your agency’s low, mid, and high scenarios, then click the blue “Save” button above the table. You will then see a table appear in the second pane displaying the saved scenarios by route. If your saved scenarios were applied to “All remaining routes”, then you can skip to Step 7.

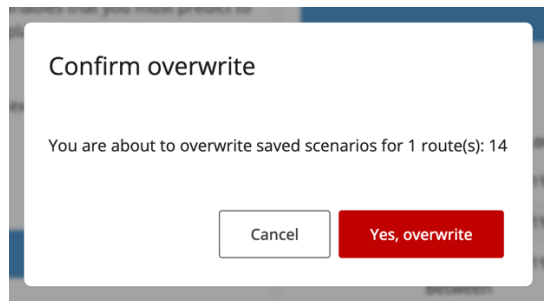
Save

**STEP 5 (CONDITIONAL):** If there are remaining routes that need change estimates, then return to Step 2, select a new route, and follow the subsequent steps.

**STEP 6 (CONDITIONAL):** If you would like to update the change estimates of a route you have already saved, you must select the “Overwrite existing routes” option at the top and follow steps 2 through 4. A pop-up will verify you are overwriting the estimates for the selected route(s).

Apply scenario to:

☐ New routes (not yet saved) ☒ Overwrite existing routes



**STEP 7:** Review the table to make sure you correctly input a scenario for each route. Then click the blue “Run Forecasts” button. You will automatically be taken to the next tab once the forecasts have been run.



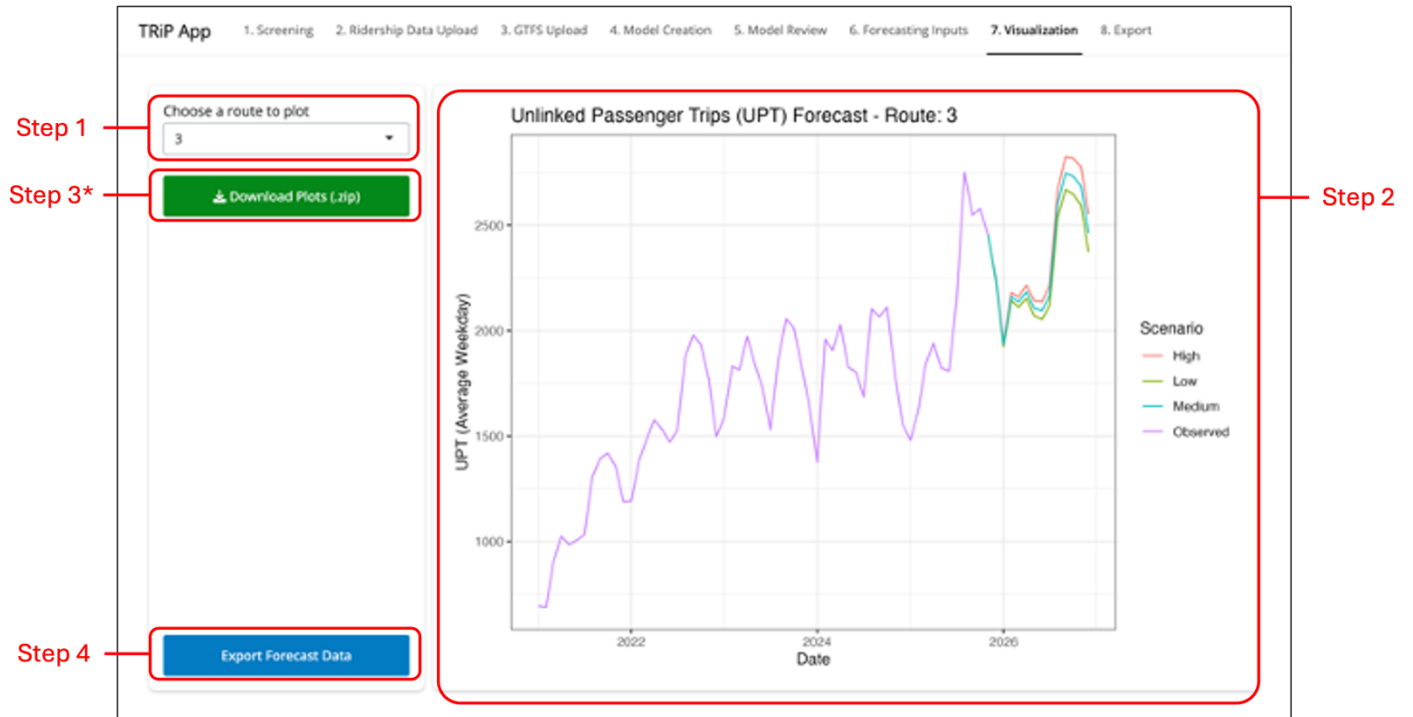
## NOTES

When updating the estimates in the cells of the table, values are in percentages, but it is not necessary to have a percentage sign.

## TAB 7: VISUALIZATION

### OVERVIEW

*This is where you may visualize the forecasts based on your selected model and your predictions.*



\* optional or conditional step

## STEPS

**STEP 1:** From the drop-down menu entitled “Choose a route to plot”, select which route you would like to visualize. Note that the last option is an option to view the summed ridership for all routes.

Choose a route to plot

all\_routes

70  
71  
75  
76  
77  
79  
all\_routes

**STEP 2:** View the selected plot showing the predicted ridership.

**STEP 3 (OPTIONAL):** Click the green “Download Plots” button to download a zipped file containing a png of each plot available.

 Download Plots (.zip)

**STEP 4:** If you would like to download the forecast data to your local computer for further viewing and analysis, click the blue “Export Forecast Data” button.

Export Forecast Data

## TAB 8: EXPORT

### OVERVIEW

*This is where you may download the results of the forecasts to your local computer.*

The screenshot displays the 'TRIP App' interface with the 'Export' tab selected. The top navigation bar includes steps from 1 to 8, with 'Export' being the active step. The main content area is divided into two panes. The left pane, labeled 'Step 2', contains instructions on downloading raw forecast data as either a CSV or Excel file, with two corresponding buttons: 'Download as CSV' and 'Download as Excel'. Below these buttons is a 'Data Description' section that defines the variables used in the forecast data. The right pane, labeled 'Step 1', displays the 'Raw Forecast Data' as a table with columns for route\_id, year, month, avg\_daily\_upt, and tot\_weekday\_upt. The table shows data for route 3 across months 1 through 8 of the year 2021. A red box highlights the 'Raw Forecast Data' table and the 'Download as CSV' and 'Download as Excel' buttons.

You can download the raw forecast data as either an Excel or a CSV file. (See a description of each of the variables below.)

**Download as CSV**

**Download as Excel**

**Data Description**

- route\_id** = Route ID
- year** = Year
- month** = Month
- avg\_daily\_upt** = Average Weekday Unlinked Passenger Trips (UPT)
- tot\_weekday\_upt** = Average Weekday UPT multiplied by total weekdays in the month.
- forecast** = True/False indicating whether the value was forecasted or not.
- scenario** = The name of the scenario that the forecasted value belongs to. If it is

**Raw Forecast Data**

Show 10 entries Search:

|   | route_id | year | month | avg_daily_upt | tot_weekday_upt |
|---|----------|------|-------|---------------|-----------------|
| 1 | 3        | 2021 | 1     | 694           | 14574.4         |
| 2 | 3        | 2021 | 2     | 688.5         | 13770           |
| 3 | 3        | 2021 | 3     | 900           | 20699           |
| 4 | 3        | 2021 | 4     | 1025.7        | 22566           |
| 5 | 3        | 2021 | 5     | 985.4         | 20694           |
| 6 | 3        | 2021 | 6     | 1005.3        | 22117           |
| 7 | 3        | 2021 | 7     | 1032.5        | 22714           |
| 8 | 3        | 2021 | 8     | 1305.6        | 28723           |

### STEPS

**STEP 1:** If desired, you may explore the raw forecast results that will appear in the second pane.

**STEP 2:** You can download the data as a .csv or an .xlsx file. Click one of the buttons in the first pane depending on your desired file type and follow the prompts in your computer's directory to choose a location and save the file.